

# Bhanu Prakash Reddy Badhiredy

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## EDUCATION

### Southern Illinois University Carbondale, IL, USA

Jan 2024 – Dec 2025

*M.S., Electrical and Computer Engineering*

**GPA 3.278**

Graduate Student Assistant in Web Applications

### Sree Vidyanikethan Engineering College, Tirupati, India

May 2023

*B.Tech., Electronics and Instrumentation Engineering*

**GPA 3.45**

## SKILLS

**Tools & scripting Languages:** ICC2, Fusion Compiler, FM and Prime Time, Caliber, Python, Perl, TCL, C Shell, JavaScript.

**Technical Skills:** PnR, Floor Planning, Clock Tree Synthesis, Routing, ERC, DRC, LVS, LEC, Physical Verification, Antenna, IR, EM, Timing Closure, DFT, Engineering Change Order, SAIF.

**STA** – Setup, Hold, POCV's, SOCV, SI Analysis, Cross-talk, High Speed Design.

**VLSI** – Register-Transfer Level (RTL) to Graphic Data System (GDS), Verilog, MOSFET, UPF, Device Physics, Sign-off convergence.

## PROFESSIONAL EXPERIENCE

### Southern Illinois University Carbondale, IL, USA

Aug 2024 – Present

*Graduate Research Assistant – ECEB.*

Designed and optimized floorplan, power plan, placement, and CTS for a 28nm DTMF chip using ICC2, improving congestion and critical path slack.

- Created Floorplan and Power plan for the DTMF chip that consists of 4 macros and 1 lakh std cells using ICC2.
- Solved All design and power-related issues using ICC2 with 2 different aspect ratios.
- Initiated Placement and CTS for a block that consists of 4 RAM and 1 lakh std cells using ICC2
- Analyzed Critical path in congestion-driven placement and reduced slack for 28nm design.

### ACL Digital, Bengaluru, India

Feb 2023 – Dec 2023

*Physical Design Engineer 1*

Executed ECOs on two tiles to resolve low-power design issues and enhance QoR. Managed block implementation from synthesis to routing, including floor planning, port placement, and PnR flow.

- Executed ECO's on 2 Tiles to resolve Low-Power design Issues and Improve Quality of Results.
- Implemented block from Synthesis to Route including Floor Planning, Port Placement, and Place and Route (PnR) Flow.
- Compiled different stages of PD Flow for 2 Tiles and interacted with Physical Verification to clear DRC's and IR Drop.
- Deployed designs in all aspects of low-power design methodologies, improved QoR.

## CERTIFICATIONS

- Salesforce developer Certification

## PROJECTS

### Identification of Hands and Wrist Movements via Surface Electromyography:

- Utilized deep neural networks to accurately identify hands and wrist movements, leading to a research paper published in the Journal of Advanced Engineering Research.
- Published a research paper titled "Identification of Hands and Wrist Movements via Surface Electromyography using Deep Neural Networks" in the Journal of Advanced Engineering Research, Volume 10, June 2023.

### Smart Plant Monitoring System:

- Developed an innovative system for monitoring plant health using sensor technology.
- Implemented data analysis algorithms to provide real-time feedback on plant conditions.

*Eligibility:* (F1-Visa) Eligible to work in the U.S. for internships and full-time employment for up to 36 months without sponsorship